Ans. to the ques. no – 1

There are two types of law in photo-chemistry such as –

① Grotthuss – Draper law : This law is defined as only those radiation which are absorbed by a reacting substance or system are responsible for producing chemical change.

This law also state that all light radiation are not bringing the chemical ~~reacting~~ reaction. Among these some are increase the kinetic energy of molecule while some are re-emitted i'e. flourescence.

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(2) Einstien law of photochemistry :-

This law also called photochemical equivalence law. This law is defined as when an atom or molecule absorbs light of a given frequency, it absorbs one quantum only.

For example :

$A + h\nu = A^*$ where h = plank consta.

Here 'A' absorb photon and become to excited state A^*.

The energy of 1 photon

$= h\nu$

$= h\frac{c}{\lambda}$

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Ans. to the ques. no – 2

Difference between photochemical and thermal reactions :–

Photochemical Reaction	Thermal Reaction
1. These reactions involve absorption of light radiations.	1. These reactions involve absorption or evolution of heat energy.
2. Presence of light is the primary requirement	2. These reactions can take place in light as well as in dark.
3. Rate of reaction is independent on temperature	3. Rate of reaction depends on temperature.
4. Photochemical activation is highly selective process	4. Thermal activation is not selective in nature
5. ΔG may be positive (+ve) or negative (–ve).	5. ΔG is always negative (–ve)
6. Energy of radiation of wavelength 200 to 800 nm, varies from 23 to 230 Kcal / mol.	6. Energy needed is in the range of 100 to 1000 kcal/ mol.

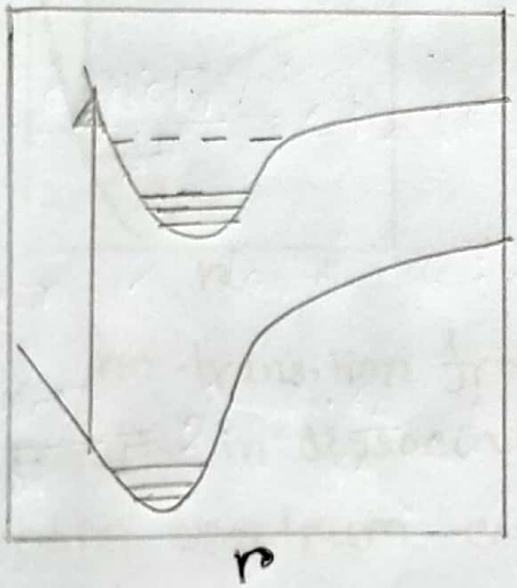
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Ans. to the ques. no-3

The primary photochemical process using molecular spectrum are given in below :-

There are four types of behaviour are of interest in connection with the primary photochemical process which are depicted in figure as follows :

Type I :

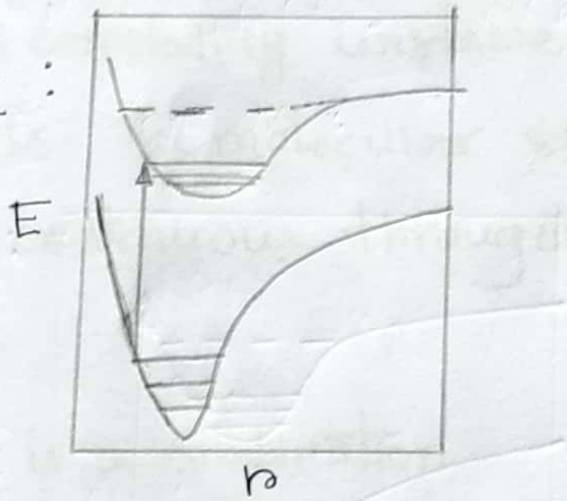


In type I, the nature of the potential energy curves is such that in the transition indicated the vibrational energy of the molecule

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in the upper electronic state exceeds the maximum value. After absorption of energy the molecule will dissociate in its 1st oscillation. In the region of discrete or banded structure, dissociation will not occur. The molecule will be in an electronically excited state.

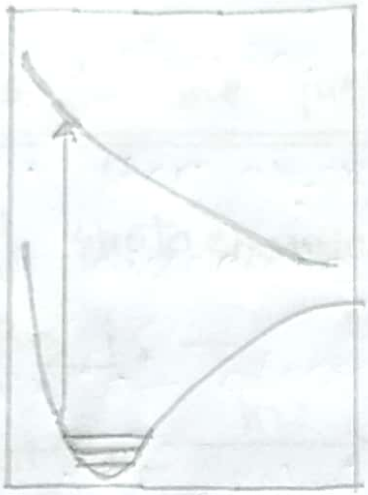
Type-II :



In type(II), no transition from lower to upper state can result in dissociateon of the molecule. The electronic spectrum consists of a series of bands with no continuous region. The primary photochemical process is then excitation of the absorbing molecule, just as in the banded region in case - I.

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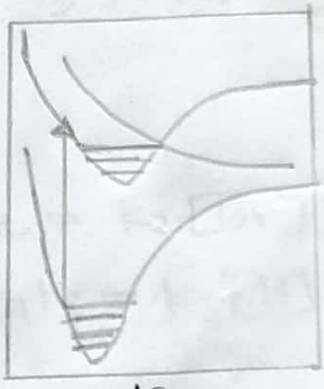
Type-III :



This type is uncommon, the upper electronic state is completely unstable and has no vibrational levels. The molecular spectrum has no bands, but is continuous throughout.

Type-IV :

This type is a combination of type-II and type-III.



There are two upper electronic levels close together; one is stable (type II) and another is unstable (type-III). This type is also called predissociation.

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Ans. to the ques: no - 5

Fluorescence :- fluorescence is defined as the emission of light by a substance that has absorbed electromagnetic radiation. when a beam light is incident on certain substances, they emit visible light or radiations and they stop emitting light or radiation as soon as the incident light is cut off and this phenomenon is known as flouresecence.

Sensitized fluoresceence :- Sensitized fluorescence is defined as a substance which is normally non-fluorescent may be made fluorescent in the presence of

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other fluorescent. For example:

$$Hg(^1S_0) + h\nu \longrightarrow Hg^*(^3P_1)$$

$$Hg^* + Tl \longrightarrow Tl^* + Hg$$

$$Tl^* \longrightarrow Tl + h\nu'$$

Tl is non-fluorescent substance and Hg is fluorescent substance.

Phosphorescence :- It is defined "when a beam of light is incident on certain substances, they emit light continuously even after the incident light is cut off. This type of delayed fluorescence is called phosphorescence.

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Ans. to the quers. no — 6

The application of gas lasers :—

i) Helium-Neon Laser :— optical demonstration, barcode scanning, spectroscopy etc.

ii) Argon laser :- Lithography, Retinal phototherapy etc.

iii) Nitrogen laser :- Measuring air pollution, pumping of dye lasers, Scientific ~~laser~~ research etc.

iv) CO_2 laser :- welding, photoacoustic spectroscopy etc.

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The application of chemical lasers :-

i) oxygen-iodine laser :- Scientific and materials research etc.

ii) All gas-phase iodine laser :- weaponry, aerospace etc.

iii) Hydrogen fluoride laser :- ~~Use~~ Research for laser weaponry etc.

The application of dye lasers :-

Dye lasers are used to isotope separation, laser medicine, birth mark removal as well as research etc.

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Ans. to the ques. no-4

The overall photochemical reaction between H_2 and Br_2 is -

$$H_2 + Br_2 \xrightarrow{h\nu} 2HBr$$

The mechanism that has been proposed for this reaction is as follows -

1. Initiation step :-

I) $Br_2 + h\nu \xrightarrow{K_1} 2Br$; Rate, $\frac{d[Br]}{dt} = K_1 I_{abs}$

2. Propagation step :-

II) $Br + H_2 \xrightarrow{K_2} HBr + H$; Rate $= K_2[Br][H_2]$

III) $H + Br_2 \xrightarrow{K_3} HBr + Br$; Rate $= K_3[H][Br]$

3. Inhibition step :-

IV) $H + HBr \xrightarrow{K_4} H_2 + Br$; Rate $= K_4[H][HBr]$

4. Termination step :-

V) $Br + Br \xrightarrow{K_5} Br_2$; Rate $= K_5[Br]^2$

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The overall rate of formation of HBr is

$$\frac{d[HBr]}{dt} = k_2[Br][H_2] + k_3[H][Br_2] - k_4[H][HBr] \quad \text{---------- (1)}$$

Applying steady state principal for H-atom and Br atom, we get

$$\frac{d[H]}{dt} = k_2[Br][H_2] - k_3[H][Br_2] - k_4[H][HBr]$$

And

$$= 0 \quad \text{---------- (II)}$$

$$\frac{d[Br]}{dt} = k_1 I_{abs} - k_2[Br][H_2] + k_3[H][Br_2] + k_4[H][HBr]$$

$$- k_5[Br]^2 = 0 \text{ ---- (3)}$$

Adding equation (2) & (3), we have

$$k_1 I_{abs} - k_5[Br]^2 = 0$$

$$\therefore [Br] = \sqrt{\frac{k_1}{k_5} \cdot I_{abs}} \quad \text{---------- (4)}$$

From equation (2), we get

$$[H] = \frac{k_2[Br][H_2]}{k_3[Br] + k_4[HBr]} \quad \text{---------- (5)}$$

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Using the equation (4) in (5), we get

$$[H] = \frac{k_2 \sqrt{\frac{k_1}{k_5} \cdot I_{abs} \cdot [H_2]}}{k_3[Br_2] + k_4[HBr]} \quad \text{---------- (6)}$$

From equation (1), (4) and (6), we get on simplification

$$\frac{d[HBr]}{dt} = \frac{2k_2\sqrt{k_1/k_5 \cdot I_{abs} \cdot [H_2]}}{1 + (k_4[HBr]/k_3 \cdot [Br])} \quad \text{---- (7)}$$

Equation (7) is the rate law equation for over all reaction.

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